Assignment 4

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Step 1.

	Word	FreqZipf	Concreteness	Valence
0	carjack	1.30	4.39	2.32
1	standardized	1.30	1.93	4.48
2	solicit	2.04	2.83	3.59
3	unfeeling	2.09	2.17	2.79
4	blimp	2.26	4.76	5.14
5	ejaculate	2.31	4.31	6.00
6	deviant	2.41	2.04	3.35
7	fallopian	2.47	3.58	4.59
8	deserter	2.52	3.32	3.27
9	reimburse	2.66	2.63	6.24
10	backwater	2.80	3.04	4.33
11	preferable	2.86	1.54	6.26
12	adhere	2.97	2.62	5.17
13	stunner	2.98	3.25	5.76
14	firearm	3.02	4.64	4.00
15	extinguisher	3.06	4.70	4.76
16	bosom	3.10	4.63	6.44
17	silvery	3.12	3.04	5.72
18	morgue	3.12	4.62	1.79
19	clubhouse	3.12	4.61	6.47
20	interface	3.16	2.81	5.32
21	elevation	3.22	3.79	5.67
22	treason	3.30	2.03	3.05
23	trivial	3.34	2.10	4.11
24	bombshell	3.34	3.88	5.68
25	harp	3.40	4.85	5.40
26	volley	3.41	3.59	5.43
27	secrecy	3.44	2.00	4.15
28	chug	3.46	3.68	5.20
29	worthless	3.51	1.97	1.89
30	foxtrot	3.53	4.25	5.83
31	victorious	3.60	2.25	6.89
32	fusion	3.60	3.30	5.60
33	hawk	3.66	4.93	6.46
34	waiter	3.66	4.67	5.05
35	moat	3.69	4.69	5.19
36	suspension	3.74	3.17	4.35
37	absurd	3.75	1.64	4.45
38	jumper	3.84	4.03	4.55
39	mature	4.05	1.86	5.95
40	compensation	4.11	3.25	6.55
41	astonishing	4.13	1.68	6.53
42	boring	4.44	2.07	2.71
43	talent	4.58	2.19	6.30
44	product	4.65	4.17	5.50
45	speech	4.72	3.37	5.62
46	victory	4.72	2.55	7.59
47	careful	4.84	1.86	6.32
48	brain	4.84	4.69	6.22
49	difficult	5.40	2.21	3.78

```
count    50.000000
mean      3.372800
std       0.876007
min       1.300000
25%       2.972500
50%       3.370000
75%       3.747500
max       5.400000
Name: FreqZipf, dtype: float64
```

From the provided dataset 50 English words were selected using sampling across five frequency bins based on the Zipf scale, giving balanced representation of both rare and frequent words.
A LLM was prompted to rate all 50 words on Concreteness and Valence using the same scales as the human norms, collected in a single batch prompt in a fresh chat session.
Generally, LLM scores strongly correlated with human norms, but the scores were slightly lower for low-frequency words, consistent with the hypothesis that LLMs depend on the frequency of word use during learning.

Step 2.

	Word	FreqZipf	Concreteness	Valence	Concreteness_LLM	Valence_LLM
0	carjack	1.30	4.39	2.32	4	2
1	standardized	1.30	1.93	4.48	2	5
2	solicit	2.04	2.83	3.59	2	4
3	unfeeling	2.09	2.17	2.79	2	2
4	blimp	2.26	4.76	5.14	6	5
5	ejaculate	2.31	4.31	6.00	4	4
6	deviant	2.41	2.04	3.35	2	2
7	fallopian	2.47	3.58	4.59	5	5
8	deserter	2.52	3.32	3.27	3	2
9	reimburse	2.66	2.63	6.24	2	6

For generation LLM-based semantic norms all 50 words were submitted. The model prompt was to rate each word on two dimensions using the same scale as the human norms : Concreteness (1–7, where 1 = fully abstract and 7 = fully concrete) and Valence (1–9, where 1 = very negative, 5 = neutral, and 9 = very positive).

The results compare of model and human, for Concreteness the model assigned low scores to abstract words such as *preferable*, *trivial*, and *secrecy* (rated 1), and high scores to physically tangible words such as *blimp*, *firearm*, and *bosom* (rated 5–6), which common with human intuition. For Valence the model gave results like *worthless* and *treason* as highly negative (1), and *victorious*, *talent*, and *victory* as highly positive (8), also consistent with human norms. These showed us that the LLM also captures the general semantic structure of the words with a reasonable basis for comparison with human ratings.

Step 3.

```
=== Correlation with FreqZipf ===
Concreteness (human): -0.1132025579454024
Concreteness (LLM): -0.15981469415990565
Valence (human): 0.3553247610642474
Valence (LLM): 0.33720431149408103

Median FreqZipf: 3.37
Low freq: 25 words, High freq: 25 words

=== Correlation Human vs LLM ===
Concreteness - low freq: 0.912746484294574
Concreteness - high freq: 0.9585355937062057
Valence - low freq: 0.8642244591783312
Valence - high freq: 0.8764906618695224

=== Mean Absolute Error ===
Concreteness - low freq: 0.6616
Concreteness - high freq: 0.732
Valence - low freq: 0.7856000000000001
Valence - high freq: 0.8724000000000001
```

Comparing High vs Low Frequency Words

To make a comparison for the frequency the dataset was split into two groups based on the median Zipf frequency (median = 3.37): 25 low-frequency words and 25 high-frequency words.

Correlation with frequency both human and LLM norms showed a weak negative correlation (human: -0.11, LLM: -0.16), and a weak positive correlation between Valence and frequency (human: 0.36, LLM: 0.34). The pattern was almost identical for both, suggesting the LLM captures the same general relationship between frequency and semantic properties as humans.

Correlations between human and LLM norms were almost lower in the low-frequency group for both Concreteness (low: 0.91, high: 0.96) and Valence (low: 0.86, high: 0.88), showing minimal worse LLM performance for rarer words.

Mean Absolute Error was higher for high-frequency words in both Concreteness (low: 0.66, high: 0.73) and Valence (low: 0.79, high: 0.87). Overall the LLM performed well in both conditions, though the pattern in correlations is consistent with the hypothesis that LLMs are less reliable for rare words.

Step 4.

1. **How do LLM norms relate to word frequency in comparison to human norms?**
   Both norms showed a similar weak relationship with word frequency. Starting with a concreteness is a bit negatively correlated with frequency, which means that rarer

words look to be rated as more concrete. While Valence shows a weak positive correlation, means more frequent words tend to be related as more positive.

2. **Do LLMs perform worse for low-frequency words? Compare correlations and errors**
   Yes, but only marginally. Correlations between LLM and human norms were slightly lower for low-frequency words compared to high-frequency words, for Concreteness and Valence. This assumes that LLMs are less accurate with rating rare words, because these words appear less frequently in training data and therefore have less available contextual information.

3. **Why might this happen?**
   LLMs learn word meanings from the contexts in which words appear during training. It explains that, frequent words occur thousands of times across diverse contexts and give the model a stable and rich representation of their meaning. While, rare words appear infrequently and in fewer contexts, resulting in less reliability in representations.

4. **What does this mean cognitively?**
   This pattern is consistent with usage-based theories of language acquisition, according to which the depth of semantic knowledge depends on the frequency and variety of occurrence of a word. Because of that, LLMs seem to behave similarly to humans - the more common a word is, the more reliably its semantic properties can be assessed. This suggests that the semantic representations of LLM are not fully symbolic, but are formed based on the distributional experience, in many ways similar to human lexical knowledge.

Step 5.
Option 2: Perturb Words
• Modify words slightly:
• e.g., apple → applf
• Change multiple letters ensuring the word is still interpretable
• Beware: LLMs use tokenization; small changes:
o May break known tokens completely
o Or create unknown/rare tokens

To complete this task I chose to execute option 2 as it allows for controlled, quantifiable manipulation of orthographic form while keeping the word length and overall structure intact. Words were orthographically rewrote by randomly substituting letters, first one letter (LD1), then two letters (LD2), creating non-words that are orthographic neighbors of the originals (e.g. *harp* → *harg*, *harp* → *hhkp*). Also a new chat session was used for each set of twisted

words to avoid any carry-over effects. The LLM was asked to rate both sets of twisted words on Concreteness and Valence using the same scales as before.

While working with LLM and presenting twisted words it didn't refuse it but instead still produced ratings. This assumes that the model gave a try to map unfamiliar orthographic forms onto known words using similarity, rather than skip them as meaningless non-words.

The effect of perturbation was that concreteness correlations between human and LLM norms ratings decreased from 0.94 to 0.69(LD1) to 0.51(LD2). A similar pattern was seen for Valence, 0.88 -> 0.76 -> 0.35. This also suggests that the high performance on original words is partially raised by data overabundance, the model recognizes the words and recalls associated ratings rather than reasoning about their meaning.

Word frequency showed only a weak and inconsistent relationship with rating robustness, showing that frequency alone does not determine how balanced LLM ratings are under orthographic perturbation.